# Kinetics Part I

Chem 151 Lab

# Rates

Rate of a chemical reaction = change of concentration of a substance per unit of time

$$\text{Average Rate} = \frac{\Delta \text{ Concentration}}{\Delta \text{ Time}}$$

Kinetics = the study of reaction rates

$$a\text{A} + b\text{B} \rightarrow c\text{C} + d\text{D}$$

$$Rate = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = +\frac{1}{c}\frac{\Delta[C]}{\Delta t} = +\frac{1}{d}\frac{\Delta[D]}{\Delta t}$$

This Week: $Rate = k[H_2O_2]^m[I^-]^n$

$$Rate = \frac{\Delta[\,]}{\Delta t} \Rightarrow PV = nRT$$

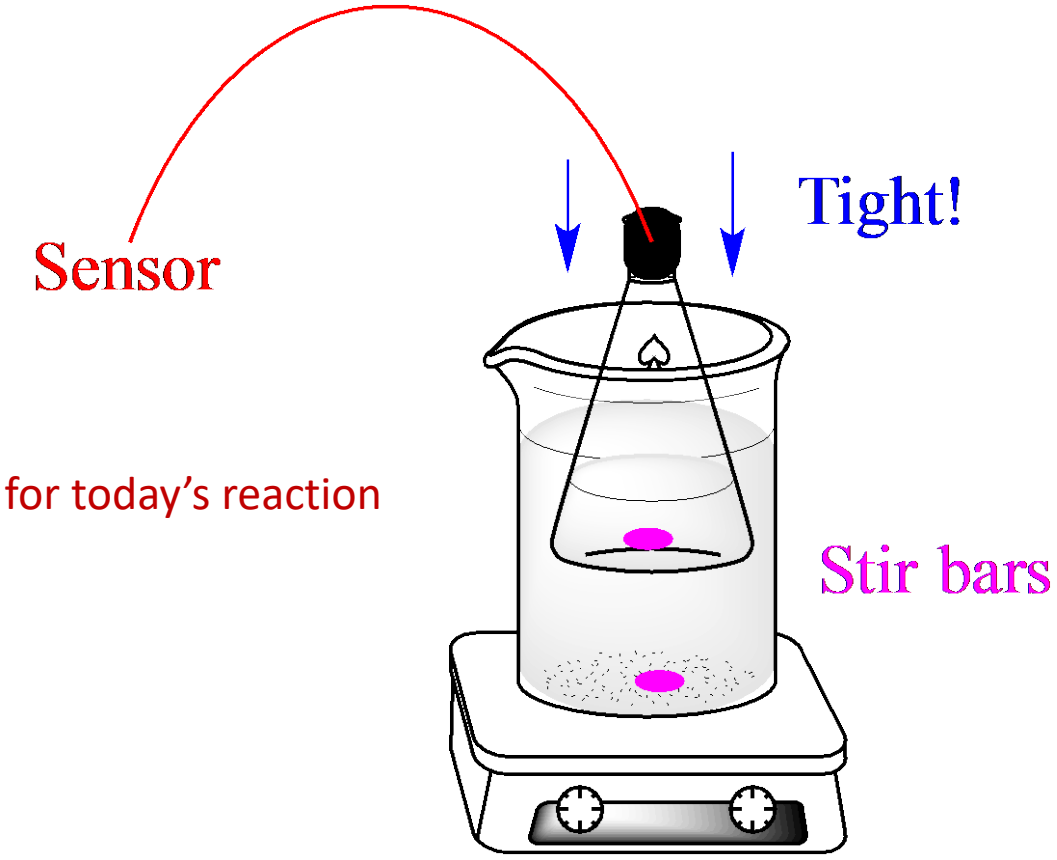
$$[\,] = \frac{n}{V} = \frac{P}{RT}$$

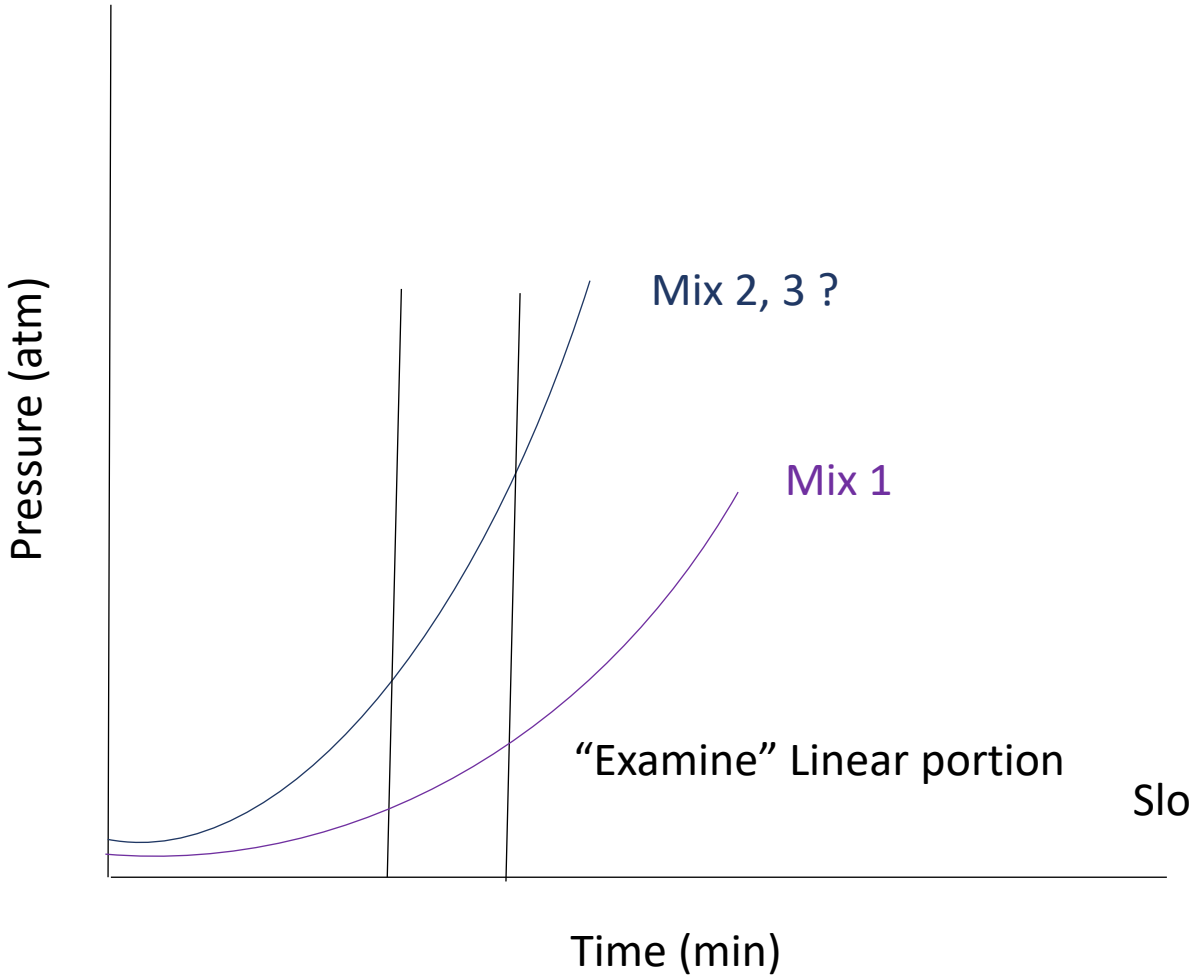
Both constant for today's reaction

$$Rate = \frac{\Delta P}{\Delta t}$$

For us today!

$$2\,H_2O_2(aq) \xrightarrow{I^-} 2\,H_2O(l) + O_2(g)$$






| | Mix 1 | Mix 2 | Mix 3 |
|---|---|---|---|
| 0.10 M KI (mL) | 15.0 | 15.0 | 30.0 |
| DI $H_2O$ (mL) | 35.0 | 25.0 | 20.0 |
| 0.88 M $H_2O_2$ (mL) | 10.0 | 20.0 | 10.0 |
| Total volume (mL) | 60.0 | 60.0 | 60.0 |

Slope = $\frac{\Delta P}{\Delta t} = rate$

$$\frac{Rate\ 2}{Rate\ 1} = \frac{Slope\ 2}{Slope\ 1} = \frac{k\{2x[H_2O_2]\}^m[I^-]^n}{k[H_2O_2]^m[I^-]^n}$$

Algebra

$$\frac{Slope\ 2}{Slope\ 1} = 2^m$$

Mix 1 → Mix 2

$[H_2O_2]$ → 2x$[H_2O_2]$

Doubles!

[KI] → [KI]

Stays constant

Mix 1 → Mix 3

$$M_1V_1 = M_2V_2$$

| | Mix 1 | Mix 2 | Mix 3 |
|---|---|---|---|
| 0.10 M KI (mL) | 15.0 | 15.0 | 30.0 |
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For example, [KI] in mix 1:

$$M_2 = \frac{M_1V_1}{V_2} = \frac{0.10M\ (15.0\ mL)}{60.0\ mL}$$

Total Volume

$$Rate = k[H_2O_2]^m[I^-]^n$$

You will calculate k (with proper units) for each of the mixes!

- Make sure all waste gets into the proper waste containers in the hood
  - Loosely capped

# Reminders

- Be sure to write a conclusion and turn your lab notebook in for Part I
- Next week is our last lab of the semester.
- You Lab Practical Final Exam will be during the last week of classes and is worth 75 points!